

Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

**Problem 1.** LCLS Parameters 20 pts.

Consider the parameters of LCLS:

| Parameters               | Values              |
|--------------------------|---------------------|
| Beam energy              | 14.35 GeV           |
| Peak current             | 3500 A              |
| Pulse length             | 230 fs              |
| FEL parameter            | $5 \times 10^{-4}$  |
| RMS angular divergence   | $1.7 \mu\text{rad}$ |
| Undulator wavelength     | 3 cm                |
| Undulator parameters $K$ | 3.7                 |

Calculate the following:

- 1) Radiation wavelength,  $\lambda_r$ .
- 2) Bunch charge of each electron bunch.
- 3) Approximate number of electrons in one micro-bunch.
- 4) 1-D gain length,  $L_{1D}$ .
- 5) Approximate undulator saturation length.
- 6) Approximate peak radiation power.
- 7) The width of the frequency spectrum at saturation.

P044 SOLUTION: LCLS Parameters 20 pts.

- 1) Radiation wavelength:

$$\lambda_r = \frac{\lambda_u}{2\gamma^2} \left( 1 + \frac{K^2}{2} \right)$$

For  $\lambda_u = 3 \text{ cm}$ ,  $\gamma = 28082$ ,  $K = 3.7$ ,  $\lambda_r \approx 0.149 \text{ nm}$ .

- 2) Bunch charge:

$$q_B = I\tau_B$$

For  $I = 3500 \text{ A}$  and  $\tau_B = 230 \text{ fs}$ ,  $q_B \approx 0.805 \text{ nC}$ .

3) Number of particles per microbunch:

$$N_{e^-} = \frac{\text{Particles per bunch}}{\text{Microbunches per bunch}} = \frac{|q_B|}{|e|} \frac{\lambda_r}{\tau_{BC}} \approx 1.1 \times 10^4$$

4) 1D gain length:

$$L_{1D} = \frac{\lambda_u}{4\pi\sqrt{3}\rho}$$

$$L_{1D} \approx 2.757 \text{ m.}$$

5) Undulator length:

$$L_s \approx 20L_G$$

$$L_s \approx 55.1 \text{ m}$$

6) Peak radiation power:

$$\rho = \frac{\text{Radiation energy at saturation}}{\text{Electron beam energy}} = \frac{\text{Peak radiation power}}{\text{Peak beam power}}$$

$$P_{\text{peak}} \approx 25.1 \text{ GW}$$

7) Width of frequency spectrum:

$$\frac{\Delta\omega}{\omega_0} \approx \sqrt{\frac{\rho}{N_u}} = \sqrt{\rho \frac{\lambda_u}{L_s}} \approx 5.2 \times 10^{-4}$$

**Problem 2.** Pillbox Cavity 30 pts.

Consider a hollow copper pillbox cavity of radius  $a = 75/2$  cm and axial length  $L = 50$  cm. Assume the interior volume of the cavity is under vacuum.

- 2 pts: What is the resonant frequency of the fundamental  $\text{TM}_{010}$  mode?
- 6 pts: Calculate the transit time factor for a particle with  $\beta = 0.85$ .
- 5 pts: Explain how the cavity operating at the fundamental frequency might be modified to increase the acceleration efficiency (larger  $T$  for given  $\beta$  found in part c). Qualitative only.
- 15 pts: For the cavity operating at the fundamental frequency, assume an RF voltage  $V_0 = E_0 d = 500$  kV and assume the cavity is made of copper with conductivity of  $\sigma = 5.9 \times 10^7$  S/m, calculate:

$U$  stored EM energy

$R_{\text{surf}}$  RF surface resistance

$P_{cy}$  average power lost over RF period

$Q_0$  Intrinsic Quality factor

$R_s$  shunt impedance

$\Delta E_{kin}$  Particle kinetic energy gain

Use formulas derived in the class notes. **Include units in your answers for all physical quantities.** Note:  $J_1(2.405) = 0.519$ .

- a) 2 pts: For the fundamental mode, we showed in class that:

$$f = \frac{2.405c}{2\pi a} = \frac{2.405 \cdot 3 \times 10^8 \text{ m/s}}{\pi \cdot 0.35 \text{ cm}} = 306 \text{ MHz}$$

- b) 6 pts:

$$T = \frac{\sin(\omega g/2v)}{\omega g/2v}$$

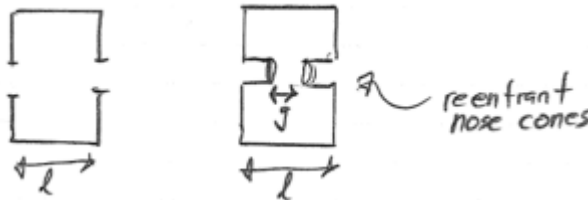
$$g = d \text{ in this case}$$

$$\omega = 2\pi f = 2\pi (306 \times 10^8 \text{ s}^{-1})$$

$$v = \beta c = 0.85c$$

$$\frac{\omega g}{2v} = \frac{\pi (306 \times 10^8 \text{ s}^{-1}) (50 \text{ cm})}{0.85c} = 1.885T = \frac{\sin(1.855)}{1.855} = \boxed{0.517}$$

- c) 5 pts: The cavity could be modified to decrease the gap length  $g$  from  $g = d$  by adding drift-tube like noes cones. This would concentrate the RF  $E_z$  over a shorter distance to increase  $T$ .



d) 15 pts:

$$\begin{aligned}
U &= \frac{1}{2} \epsilon_0 E_{010}^2 da^2 J_1^2 (2.405) \\
V_0 = E_0 d, E_0 &= \frac{500 \text{ kV}}{50 \text{ cm}} = 10 \text{ kV/cm} = 1 \text{ MV/m} \\
U &= \frac{1}{2} (8.854 \times 10^{-12} \text{ F/m}) (1 \times 10^6 \text{ V/m})^2 (0.5 \text{ m}) (0.75/2 \text{ m})^2 (0.519)^2, \boxed{U = 0.0838 \text{ J}} \\
R_{surf} &= \frac{1}{\sigma \delta_s} \\
\delta_s &= \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{2}{2\pi (306 \times 10^6 \text{ s}^{-1}) (\mu_0) (5.9 \times 10^7 \text{ S/m})}} = 3.75 \times 10^{-6} \text{ m} = 3.75 \text{ } \mu\text{m} \\
R_{surf} &= \frac{1}{(5.9 \times 10^7 \text{ S/m}) (3.75 \times 10^{-6} \text{ m})}, \boxed{R_{surf} = 0.00452 \text{ } \Omega} \\
P_{cy} &= \frac{1}{2} \pi \epsilon_0 \omega \delta_s V_0^2 J_1^2 (2.405) \frac{a(a+d)}{d^2} \\
P_{cy} &= \frac{1}{2} \pi \epsilon_0 (2\pi (306 \times 10^6 \text{ s}^{-1})) (3.75 \times 10^{-6} \text{ m}) (500 \times 10^3 \text{ V})^2 (0.519)^2 \frac{(0.75/2 \text{ m}) (0.75/2 \text{ m} + 0.5 \text{ m})}{(0.5 \text{ m})^2} \\
\boxed{P_{cy} = 8.891 \text{ KW} = 8891 \text{ W}} \\
Q_0 &= \frac{\omega U}{P_{cy}} = \frac{2\pi (306 \times 10^6 \text{ s}^{-1}) (0.0838 \text{ J})}{8891 \text{ W}}, \boxed{Q_0 = 19375 = 1.9 \times 10^4} \\
R_s &= \frac{V_{cy}^2}{2P_{cy}} \\
V_{cy} &= V_0 T = (500 \text{ kV}) (0.517) = 258.5 \text{ kV} \\
R_s &= \frac{(258.5 \text{ kV})^2}{2 (8.891 \text{ kW})}, \boxed{R_s = 3.75 \text{ M}\Omega = 3.75 \times 10^6 \text{ } \Omega} \\
\Delta E_{kin} &= |q| V_0 T = |q| V_{cy} = e V_{cy} \\
\Delta E_{kin} &= e (258.5 \text{ kV}), \boxed{\Delta E_{kin} = 258.5 \text{ keV}}
\end{aligned}$$

**Problem 3.** Synchrotron Energy Acceptance 35 pts.

Consider the continuous approximation RF focusing/acceleration equations derived in class for a ring:

$$\begin{aligned}
\frac{d}{dn} \phi &= -\frac{2\pi h \eta_s}{\gamma_s \beta_s^2} \frac{\Delta W}{mc^2} \\
\frac{d}{dn} \Delta W &= q V_{rf} (\sin \phi - \sin \phi_s)
\end{aligned}$$

with  $\eta_s = \frac{1}{\gamma_s^2} - \frac{1}{\gamma_{tr}^2}$  the usual slip factor. Assume that the coefficients (i.e.,  $\beta_s$ ,  $\eta_s$ , and  $V_{rf}$ ) remain approximately constant lap-by-lap in the evolution.

a) 10 pts: Show for a particle evolving in the bucket that

$$\frac{1}{2} \left( \frac{d\phi}{dn} \right)^2 - \frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 mc^2} (\cos \phi + \phi \sin \phi_s) = \text{const}$$

Hint: Combine to make one equation. Multiply by  $\frac{d\phi}{dn}$ . Then integrate with respect to  $dn$ .

- b) 5 pts: Show that the result in a) can be expressed as:

$$\Delta W^2 - \frac{\beta_s^2 \gamma_s m c^2 q V_{rf}}{\pi h \eta_s} (\cos \phi + \phi \sin \phi_s) = \text{const}$$

- c) 10 pts:

Choose  $\phi_s$  for maximum acceptance (corresponding to no acceleration) below transition and show that for this case that the separatrix equation is:

$$\Delta W^2 - \frac{2\gamma_s \beta_s^2 m c^2 q V_{rf}}{\pi h \eta_s} \cos^2 \left( \frac{\phi}{2} \right) = 0$$

Hint: It is easy to identify the constant at the  $x$ -points at  $\phi = \pm\pi$ .

- d) 10 pts: Apply the result in part c) to estimate the energy acceptance  $\Delta W/W$  of a proton synchrotron with:

$h = 80$  harmonic number  
 $\eta_s = 0.05$  slip factor  
 $V_{rf} = 1 \text{ MV}$  rf Voltage  
 $W = 200 \text{ MeV}$  Kinetic Energy

Assume incoming particles have the correct synchronous phase and that the machine is operated for maximum acceptance.

### Synchrotron Energy Acceptance 35 pts.

- a) 10 pts: Differentiate the phase equation and use the energy equation to obtain:

$$\frac{d^2}{dn^2} \phi = -\frac{2\pi h \eta_s}{\gamma_s \beta_s^2 m c^2} \frac{d\Delta W}{dn} = -\frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 m c^2} (\sin \phi - \sin \phi_s)$$

which we write as

$$\frac{d^2}{dn^2} \phi + \frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 m c^2} (\sin \phi - \sin \phi_s) = 0$$

Multiply by  $d\phi/dn$  and integrate:

$$\begin{aligned} \int \frac{d^2 \phi}{dn^2} \frac{d\phi}{dn} dn + \frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 m c^2} (\sin \phi - \sin \phi_s) \frac{d\phi}{dn} dn &= 0 \\ \Rightarrow \int \frac{1}{2} \frac{d}{dn} \left( \frac{d\phi}{dn} \right)^2 dn - \frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 m c^2} (\cos \phi + \phi \sin \phi_s) &= \text{const} \end{aligned}$$

Giving:

$$\frac{1}{2} \left( \frac{d\phi}{dn} \right)^2 - \frac{2\pi h \eta_s q V_{rf}}{\gamma_s \beta_s^2 m c^2} (\cos \phi + \phi \sin \phi_s) = \text{const}$$

- b) 5 pts: Substitute:

$$\frac{d\phi}{dn} = -\frac{2\pi h \eta_s}{\gamma_s \beta_s^2} \frac{\Delta W}{m c^2}$$

giving

$$\frac{1}{2} \left( \frac{2\pi\hbar\eta_s}{\gamma_s\beta_s^2 mc^2} \right)^2 \Delta W^2 - \frac{2\pi\hbar\eta_s}{\gamma_s\beta_s^2} \frac{qV_{rf}}{mc^2} (\cos\phi + \phi \sin\phi_s) = \text{const}$$

to obtain

$$\Delta W^2 - \frac{\beta_s^2 \gamma_s mc^2 qV_{rf}}{\pi\hbar\eta_s} (\cos\phi + \phi \sin\phi_s) = \text{const}$$

with a rescaled constant.

- c) 10 pts: Below transition,  $\eta_s > 0$ , expect  $\phi_s = 0$  (no acceleration) to lead to maximum focusing and acceptance. This gives:

$$\Delta W^2 - \frac{\beta_s^2 \gamma_s mc^2 qV_{rf}}{\pi\hbar\eta_s} \cos\phi = \text{const}$$

On separatrix X-points,  $\Delta W = 0$ ,  $\phi = \pm\pi$ , so the constant is identified as

$$\text{const} = \frac{\gamma_s \beta_s^2 mc^2 qV_{rf}}{\pi\hbar\eta_s}$$

Using this result, the constraint equation becomes

$$\Delta W^2 - \frac{\gamma_s \beta_s^2 mc^2 qV_{rf}}{\pi\hbar\eta_s} (1 + \cos\phi) = 0$$

or equivalently, using  $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$ , we have

$$\Delta W^2 - \frac{2\gamma_s \beta_s^2 mc^2 qV_{rf}}{\pi\hbar\eta_s} \cos^2 \left( \frac{\phi}{2} \right) = 0$$

- d) 10 pts: Solve for  $\Delta W$ :

$$\Delta W = \pm \sqrt{\frac{2\gamma_s \beta_s^2 mc^2 qV_{rf}}{\pi\hbar\eta_s}} \cos \left( \frac{\phi}{2} \right)$$

But  $\phi = 0$  for the max  $\Delta W$  giving

$$\Delta W = \pm \sqrt{\frac{2\gamma_s \beta_s^2 mc^2 qV_{rf}}{\pi\hbar\eta_s}}$$

Evaluate this expression using the given numbers

$$mc^2 = 938 \text{ MeV for protons}$$

$$h = 80$$

$$qV_{rf} = eV_{rf} = 1 \text{ MeV}$$

$$\eta_s = 0.05$$

and the relativistic factors

$$\gamma_s = \frac{W + mc^2}{mc^2} = \frac{200 \text{ MeV} + 938 \text{ MeV}}{938 \text{ MeV}} = 1.213$$

$$\beta_s = \sqrt{1 - 1/\gamma_s^2} = \sqrt{1 - 1/(1.213)^2} = 0.566$$

Evaluating the formula:

$$\Delta W = \pm \sqrt{\frac{2(1.213)(0.566)^2(938)}{\pi(80)(0.05)}} \text{ MeV} = \pm 7.620 \text{ MeV}$$

Or

$$\frac{\Delta W}{W} = \pm \frac{7.620 \text{ MeV}}{200 \text{ MeV}} = \pm 0.0381 = \pm 3.81\%$$